Layout

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Outline

- nch vs nmos_rf
- Layout
- DRC
- LVS
- PEX
- Comparison
- Summary

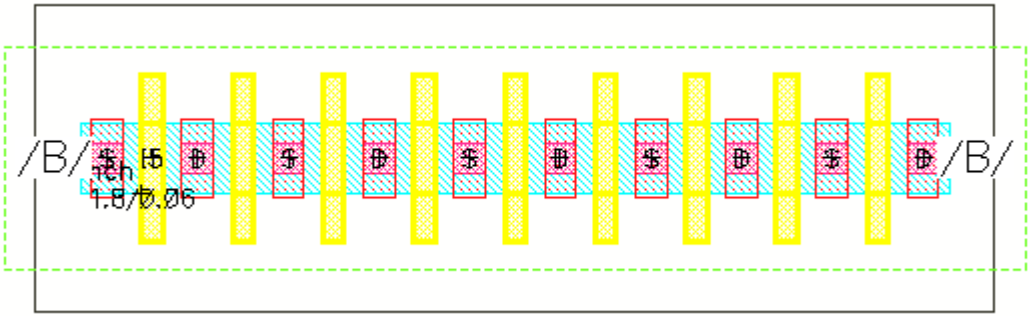
nch vs nmos_rf

구분	nch (Standard)	nmos_rf (RF)
주요 용도	디지털 로직, 저주파 Analog	LNA, Mixer, VCO 등 RF 회로
모델링 정밀도	DC 및 저주파 위주	GHz 대역 고주파 특성 포함
Layout	최소 면적	Guard Ring 포함
Gate 저항	대략적 모델링	정밀 모델링
Deep N-well	Bulk가 공통 P-sub에 연결되는 경우가 많아, 거의 불필요	기판을 통해 들어오는 다른 회로의 노이즈를 차단하기 위해, NMOS의 Bulk를 독립적인 노드로 분리하는 경우가 많음

nch vs nmos_rf

	nch				nmos_rf		
FoM	154.7271439	154.2480439	154.1818546		153.8811375	154.3109795	154.6751972
|PN|	76.9	75.27	74.01		90.84335	90.51131	90.33986
fosc	1.64E+10	2.10E+10	2.68E+10		3.88E+09	4.50E+09	5.01E+09
df	1.00E+06	1.00E+06	1.00E+06		1.00E+06	1.00E+06	1.00E+06
Idc	3.683	4.65	5.766		3.736	4.223	4.625
Pdc/1mW	4.4196	5.58	6.9192		7.472	8.446	9.25
	ss	tt	ff		ss	tt	ff

nch vs nmos_rf



nch

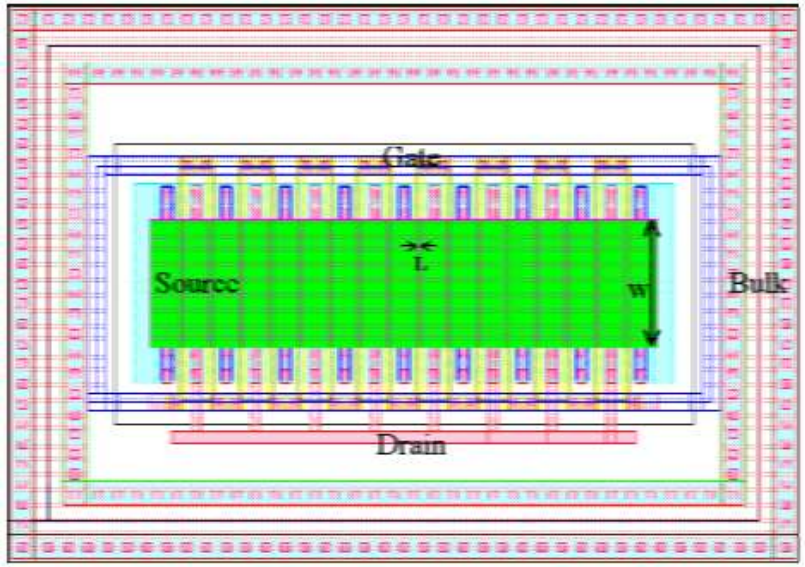


Fig. 2-1: Layout of the NMOS tester.

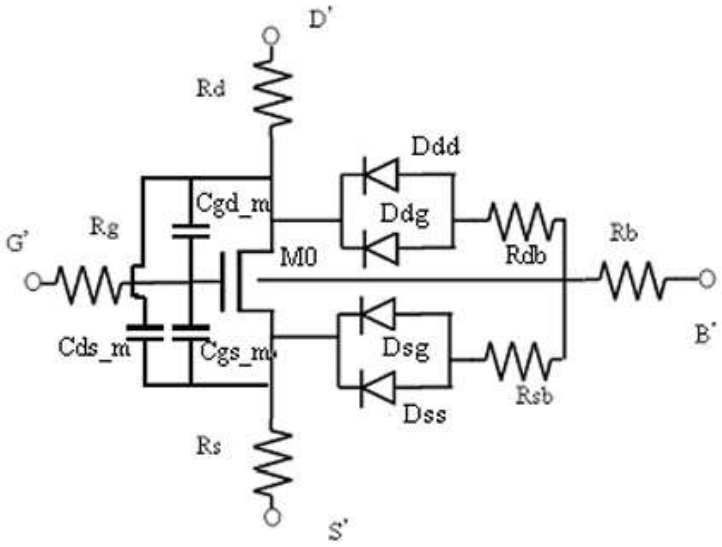
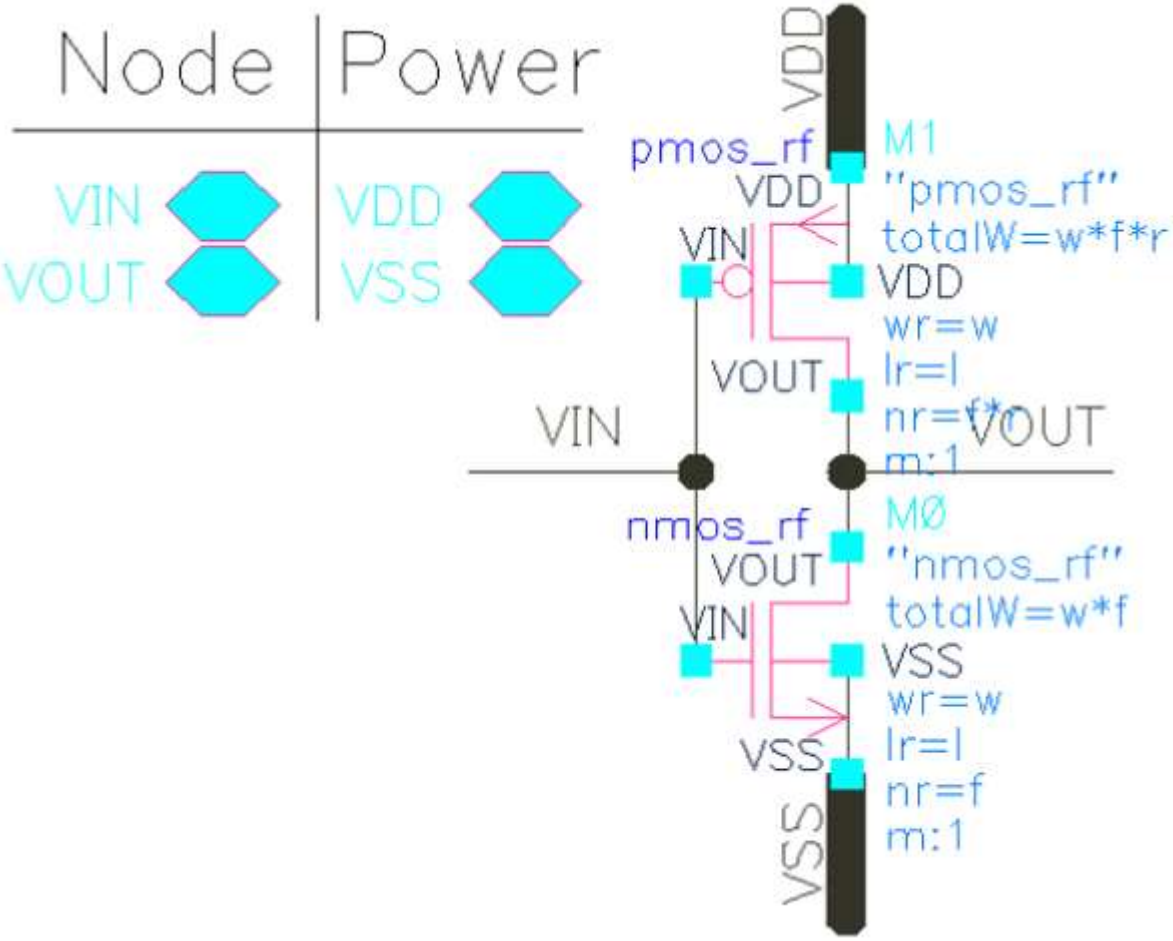
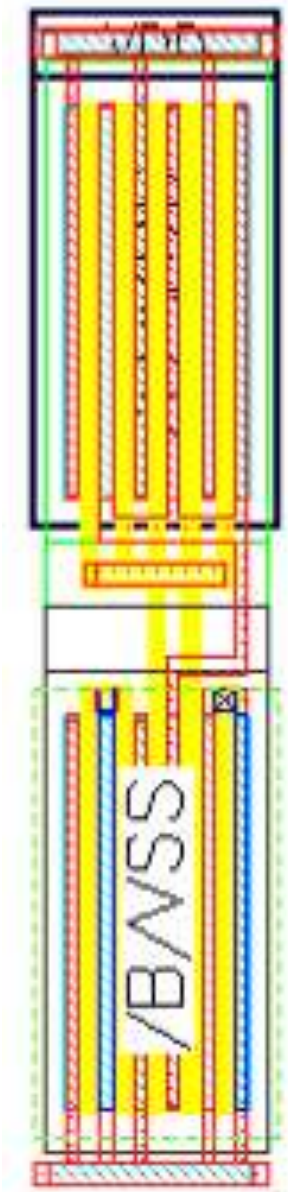


Fig. 2-2: Equivalent circuit model for the RF NMOS transistor.

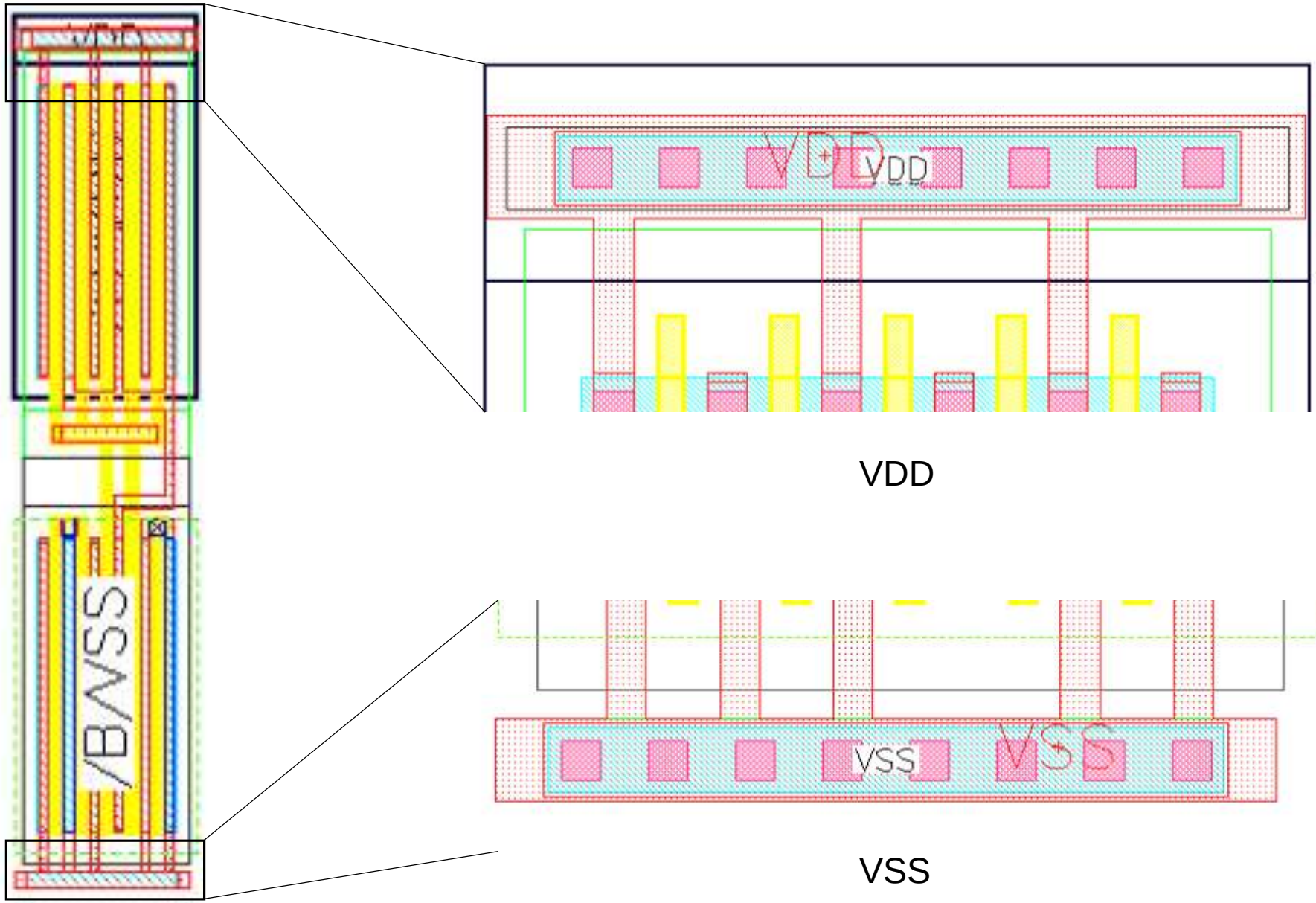
nmos_rf

Layout


Node
Power
VIN
VOUT
VDD
VSS
VDD
pmos_rf
M1
"pmos_rf"
totalW=w*f*r
VDD
VIN
VDD
wr=w
lr=l
VOUT
nr=f*r
VIN
VOUT
m:1
nmos_rf
MØ
"nmos_rf"
VOUT
totalW=w*f
VIN
VSS
wr=w
lr=l
VSS
nr=f
m:1
VSS


/B/VSS

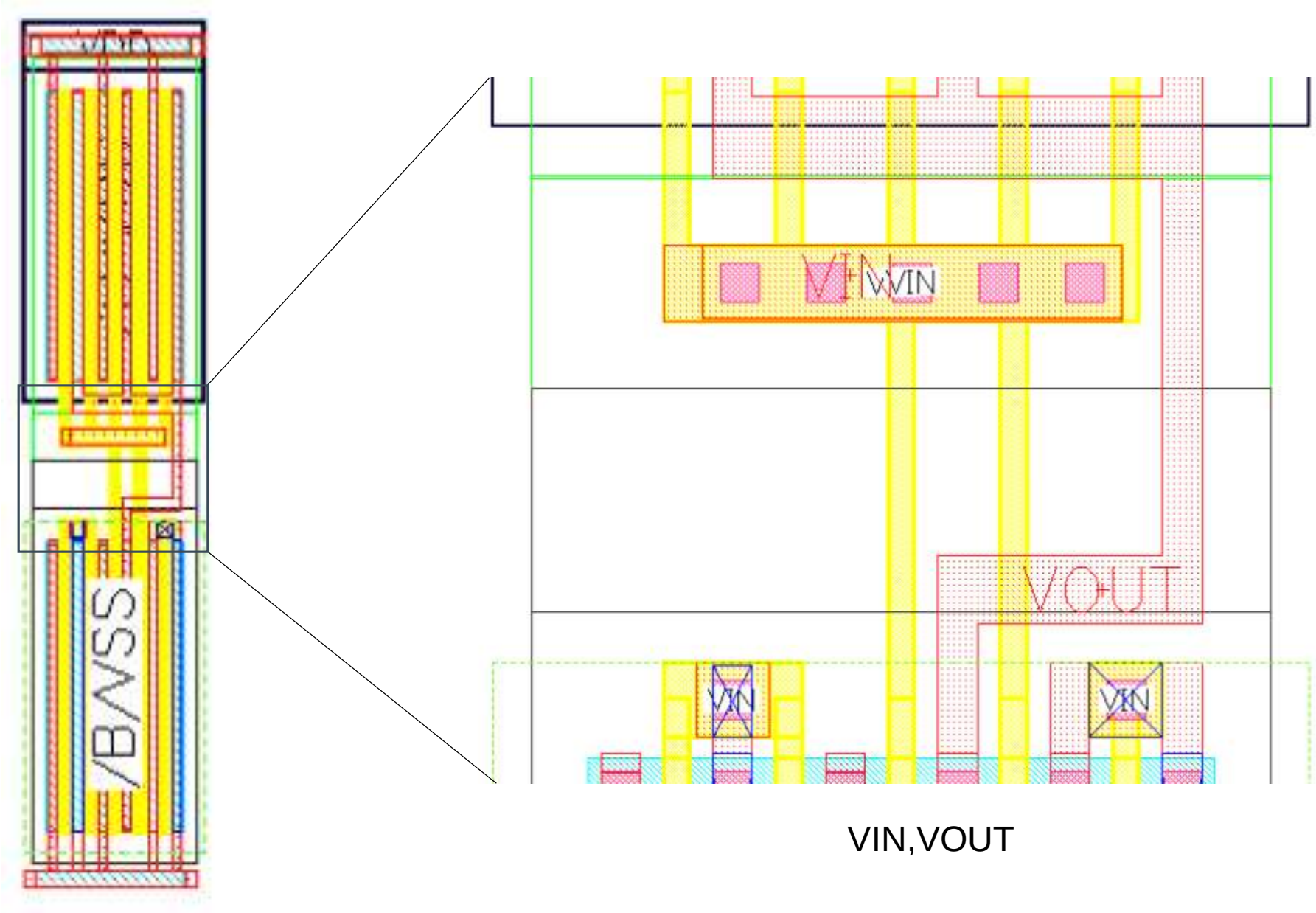
Layout



VDD

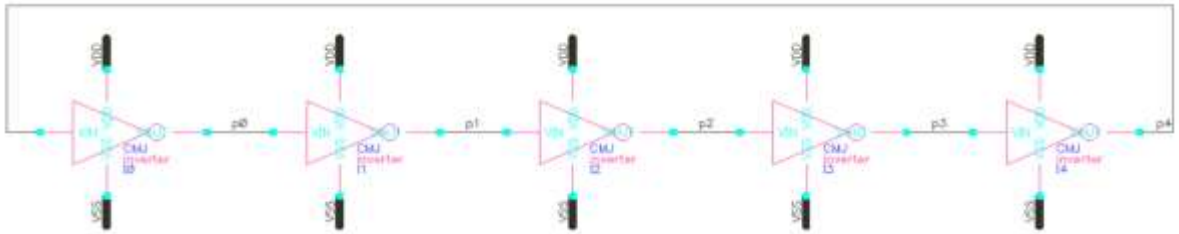
VSS

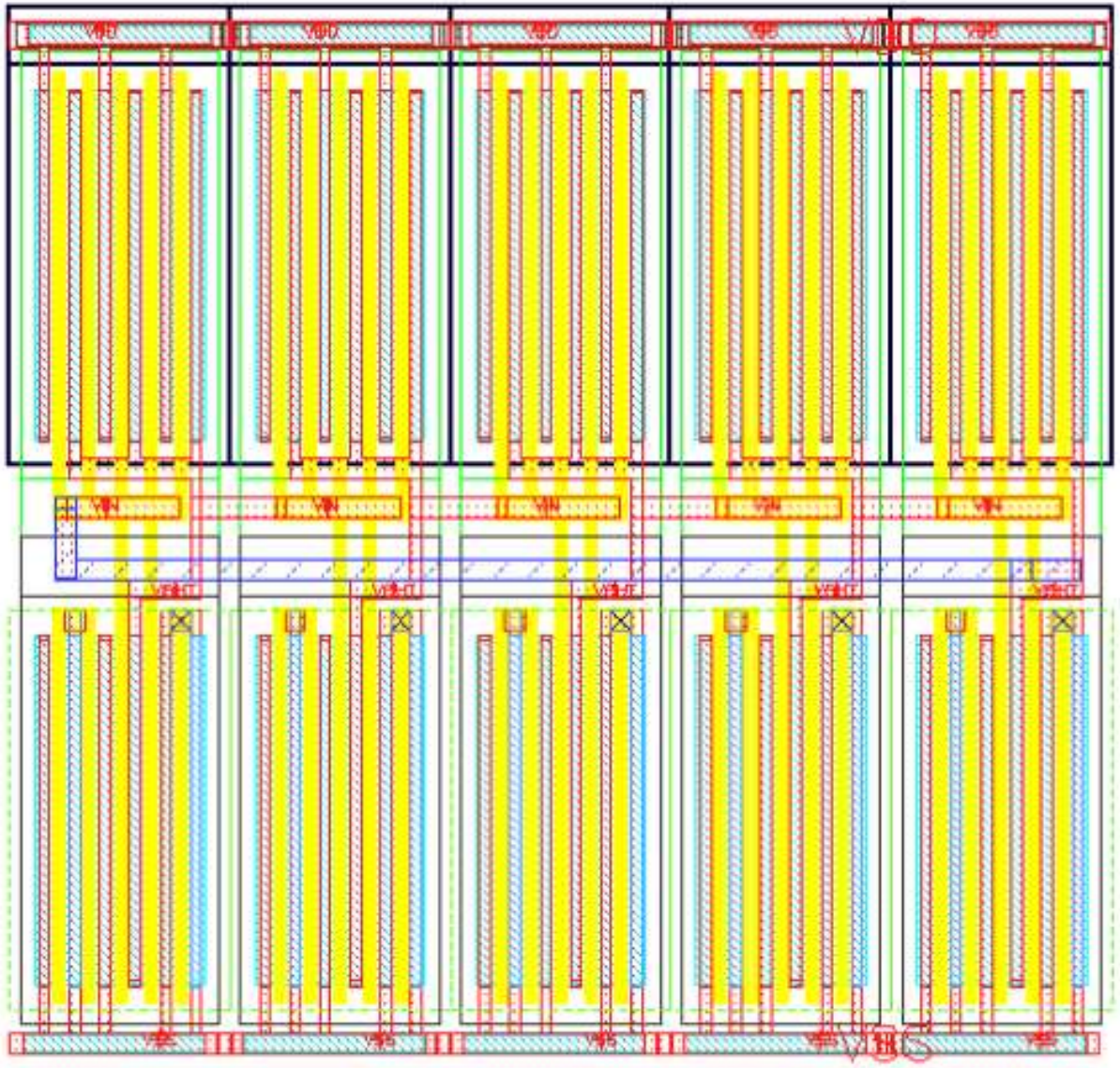
Layout



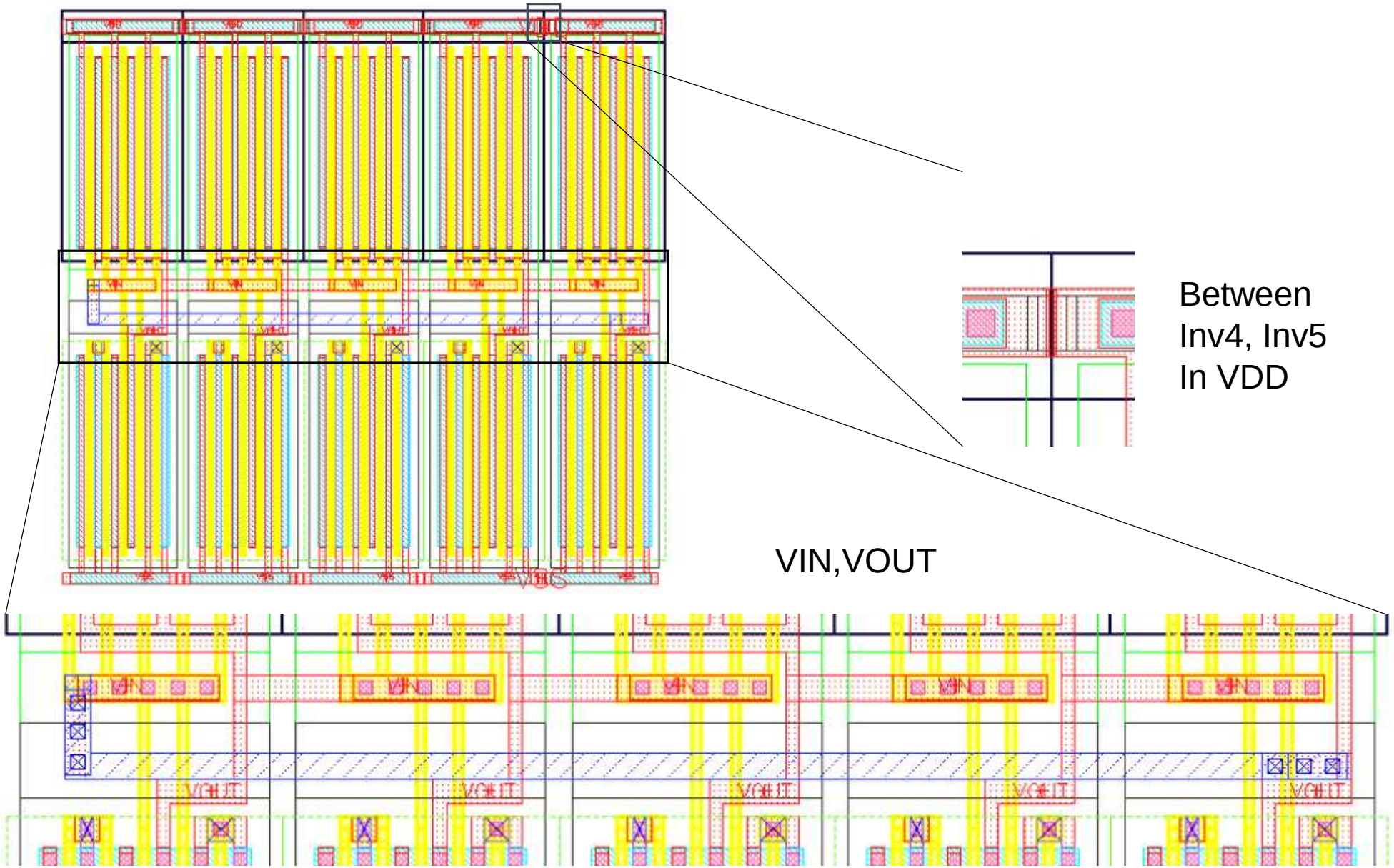
VIN,VOUT

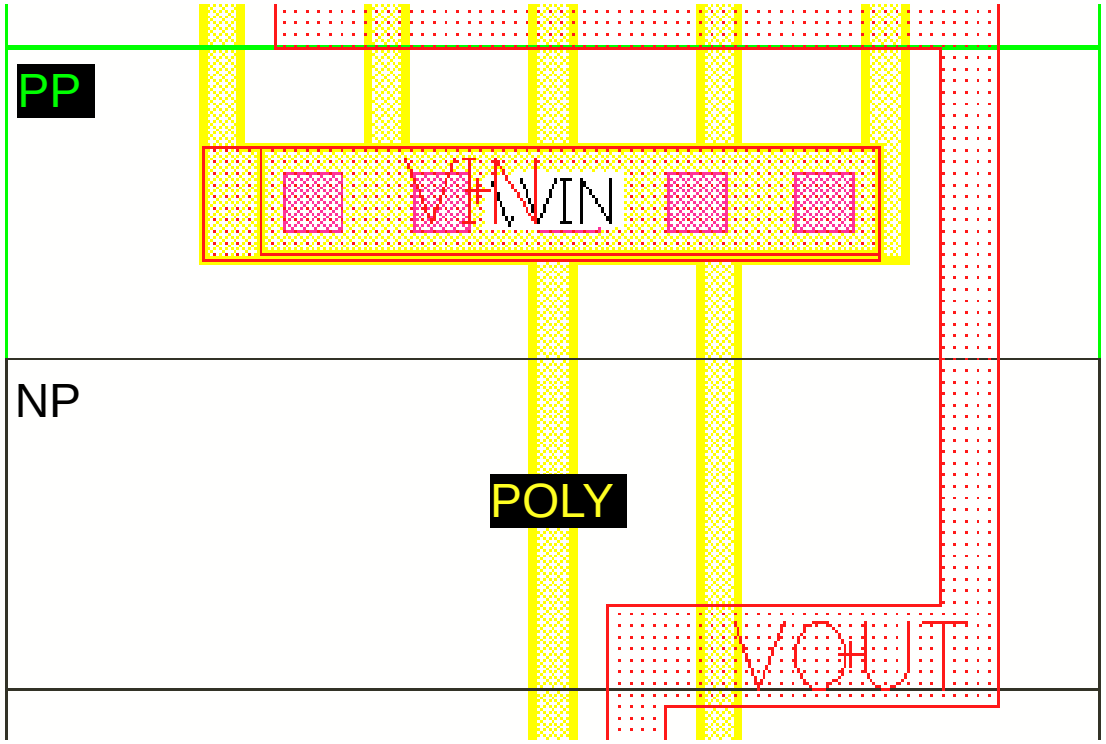
Layout





Layout



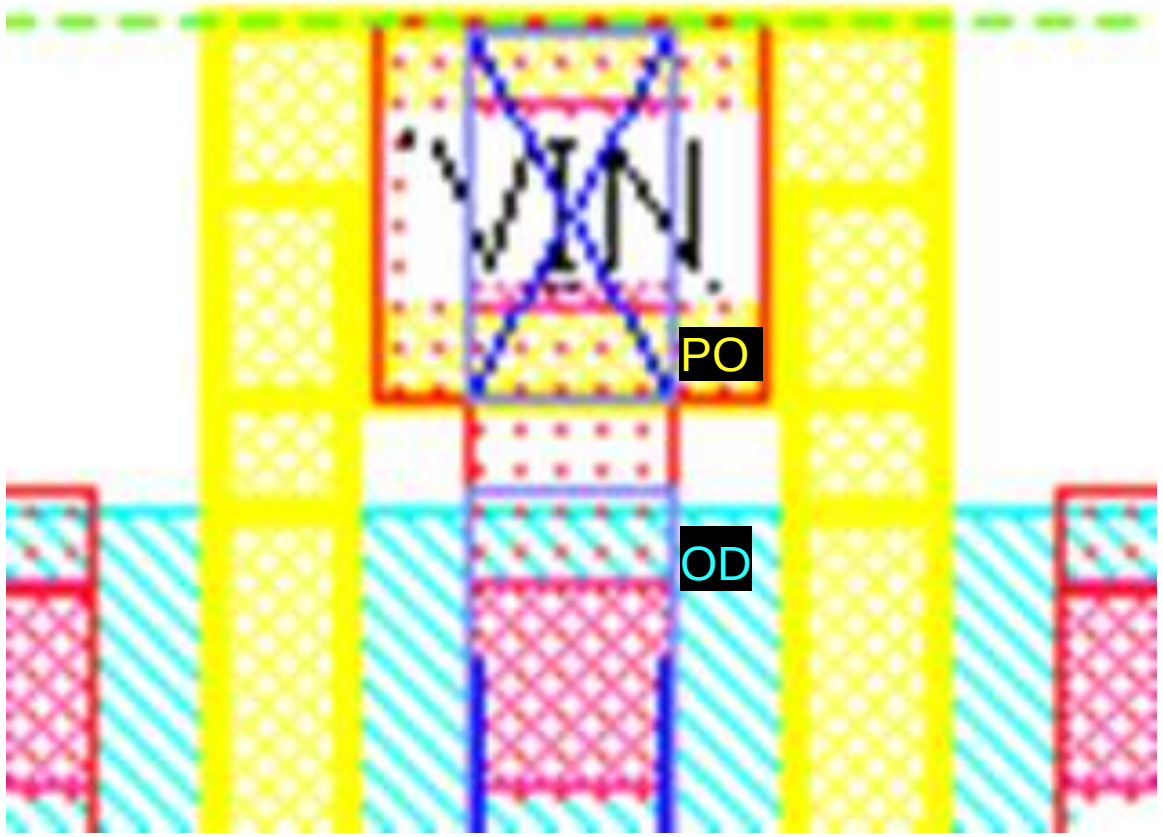


PP.EN.1 { @ {NP OR PP} enclosure of PO (except DPO) >= 0.15 um

ENC POLY IMP < PP_EN_1 ABUT < 90 SINGULAR REGION

((POLY NOT SRAM_EXCLUDE) NOT LOGO) NOT IMP

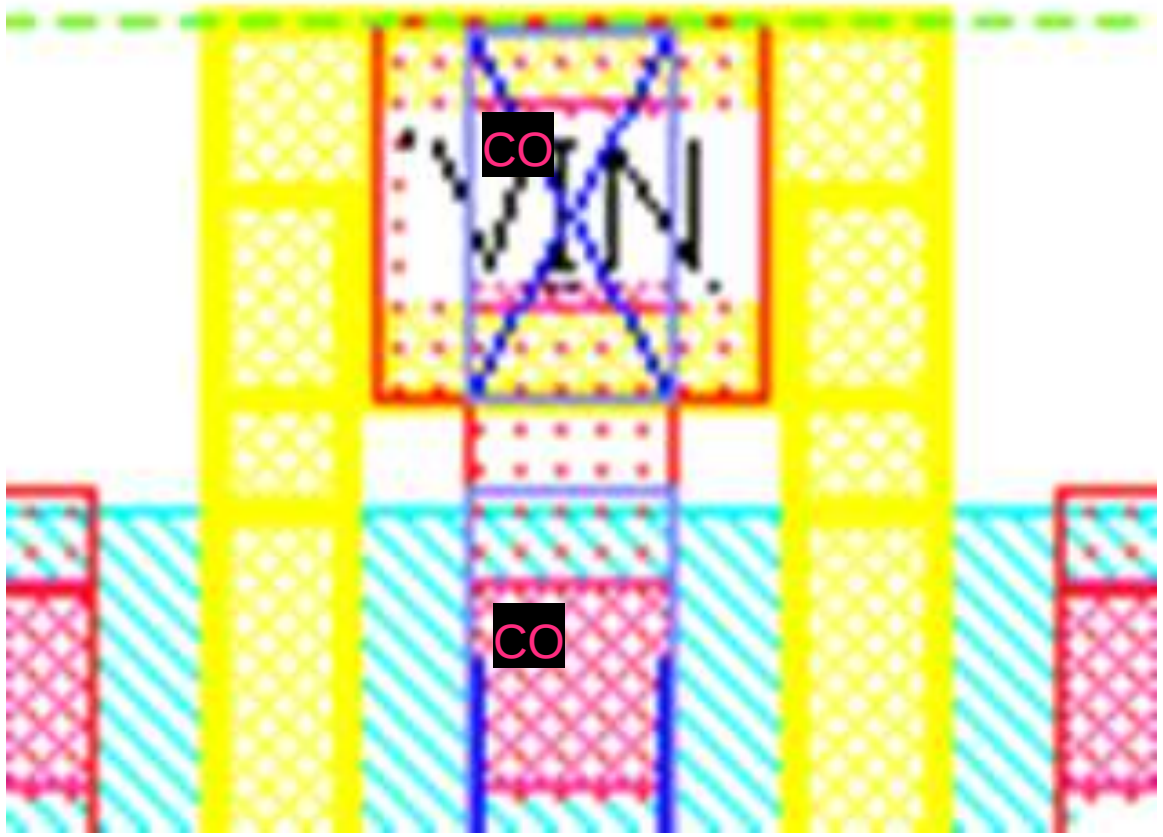
PO should be enclosed with NP or PP



PO.S.4 { @ Field PO space to OD >= 0.05 um
  EXT POLY OD < PO_S_4 ABUT < 89.5 SINGULAR REGION

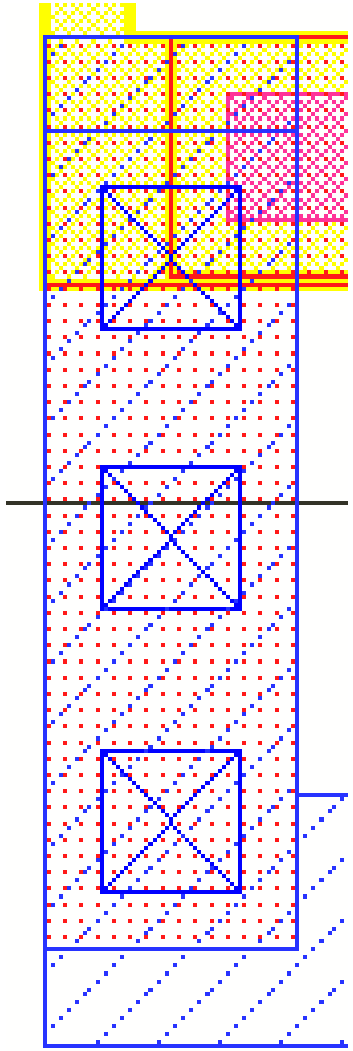
Field PO should be at
a distance of 0.05 um from OD

DRC



CO.S.1 { @ Spacing >= 0.11 um
  EXT CO < CO_S_1 ABUT < 90
SINGULAR REGION

CO should be at a distance of 0.11 um from another CO



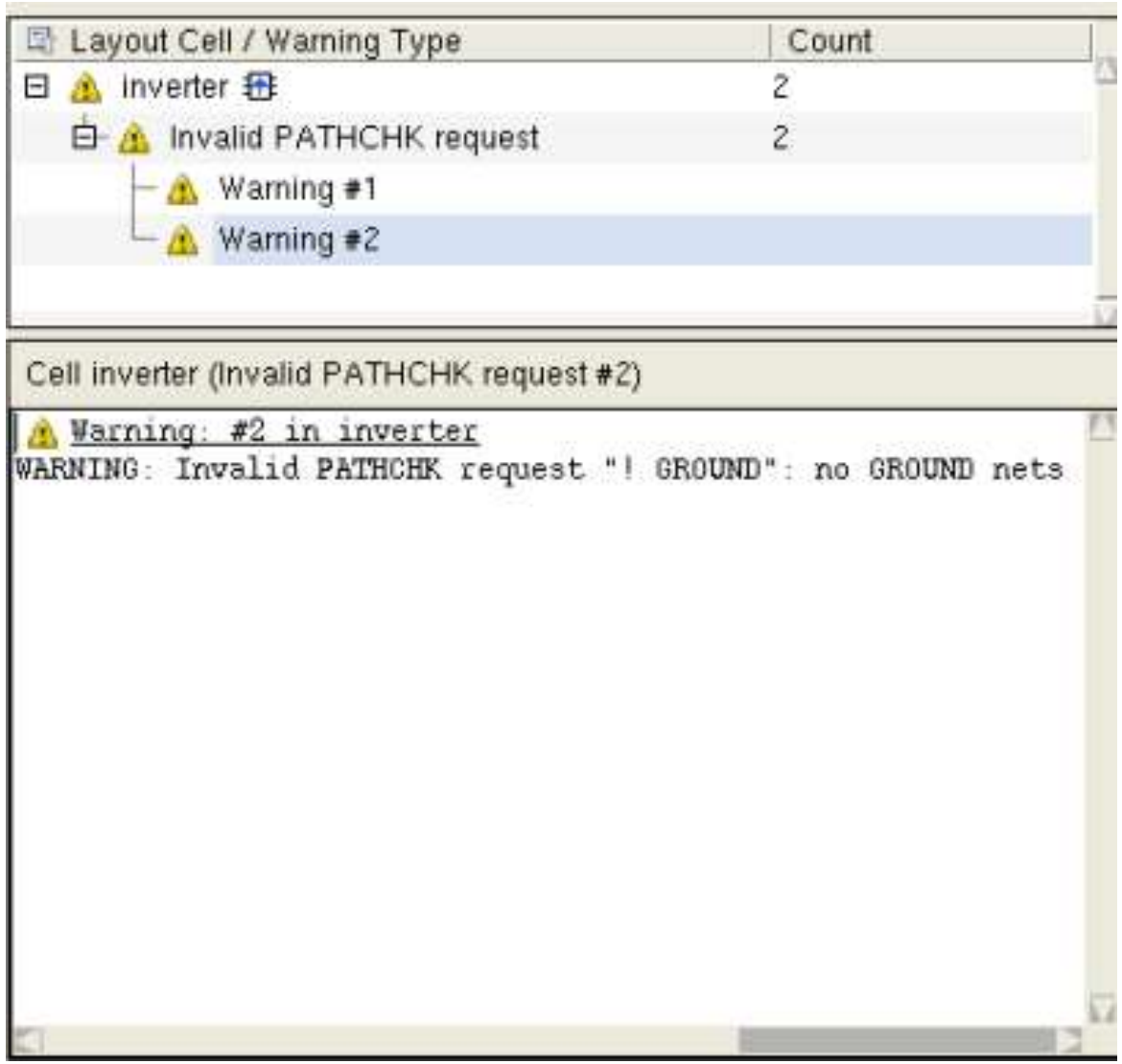
CO x3

VIA1.R.2__VIA1.R.3 { @ When M1 or M2 width > 0.3 um, more than one VIA1 is required.

~~~~~~~~~~~~~~~~~~~~~~~~~~~~

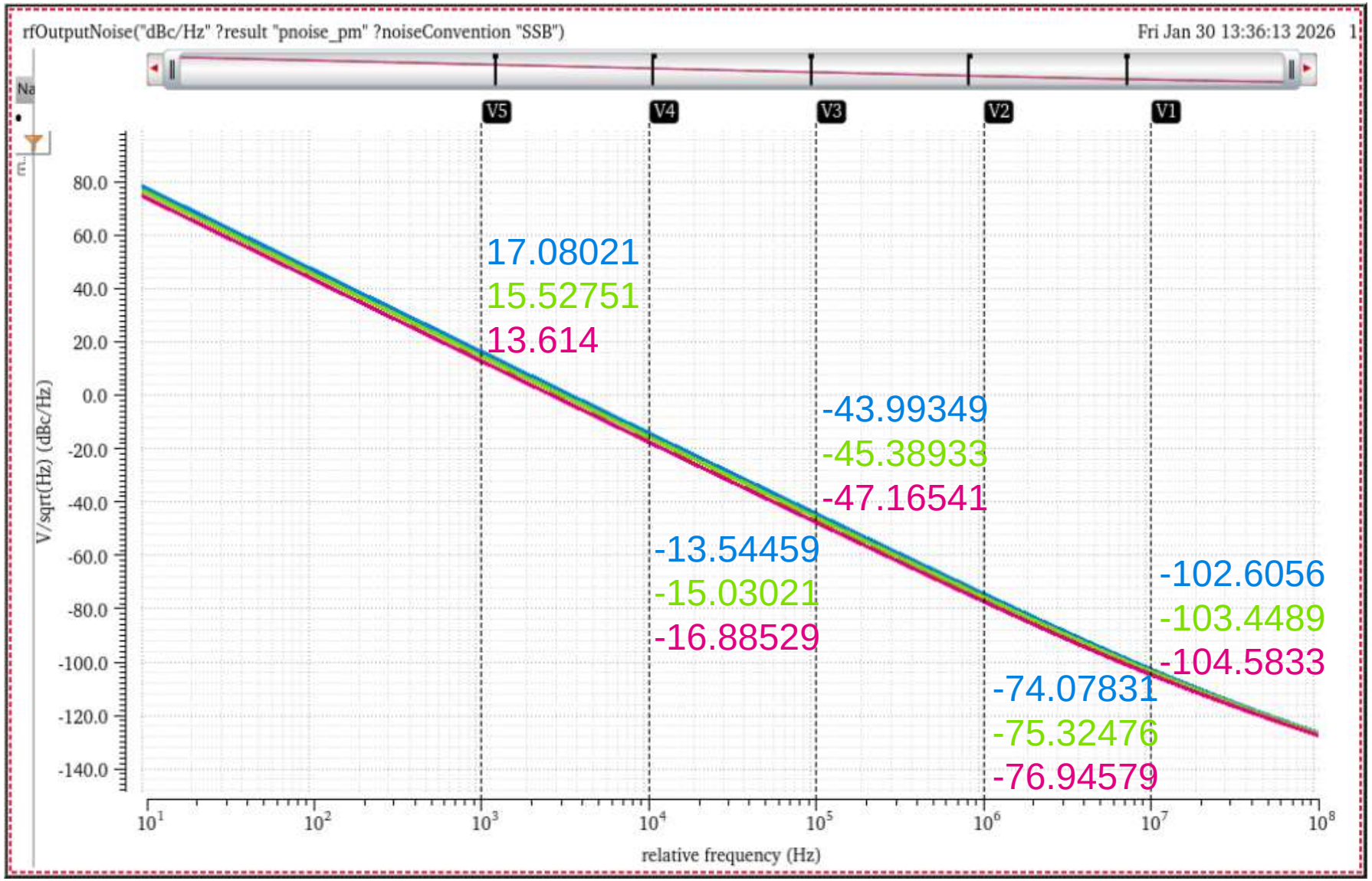
When M1 or M2 width is larger than 0.3um, more than one VIA1 is required.
~~~~~~~~~~~~~~~~~~~~~~~~~~~~

LVS



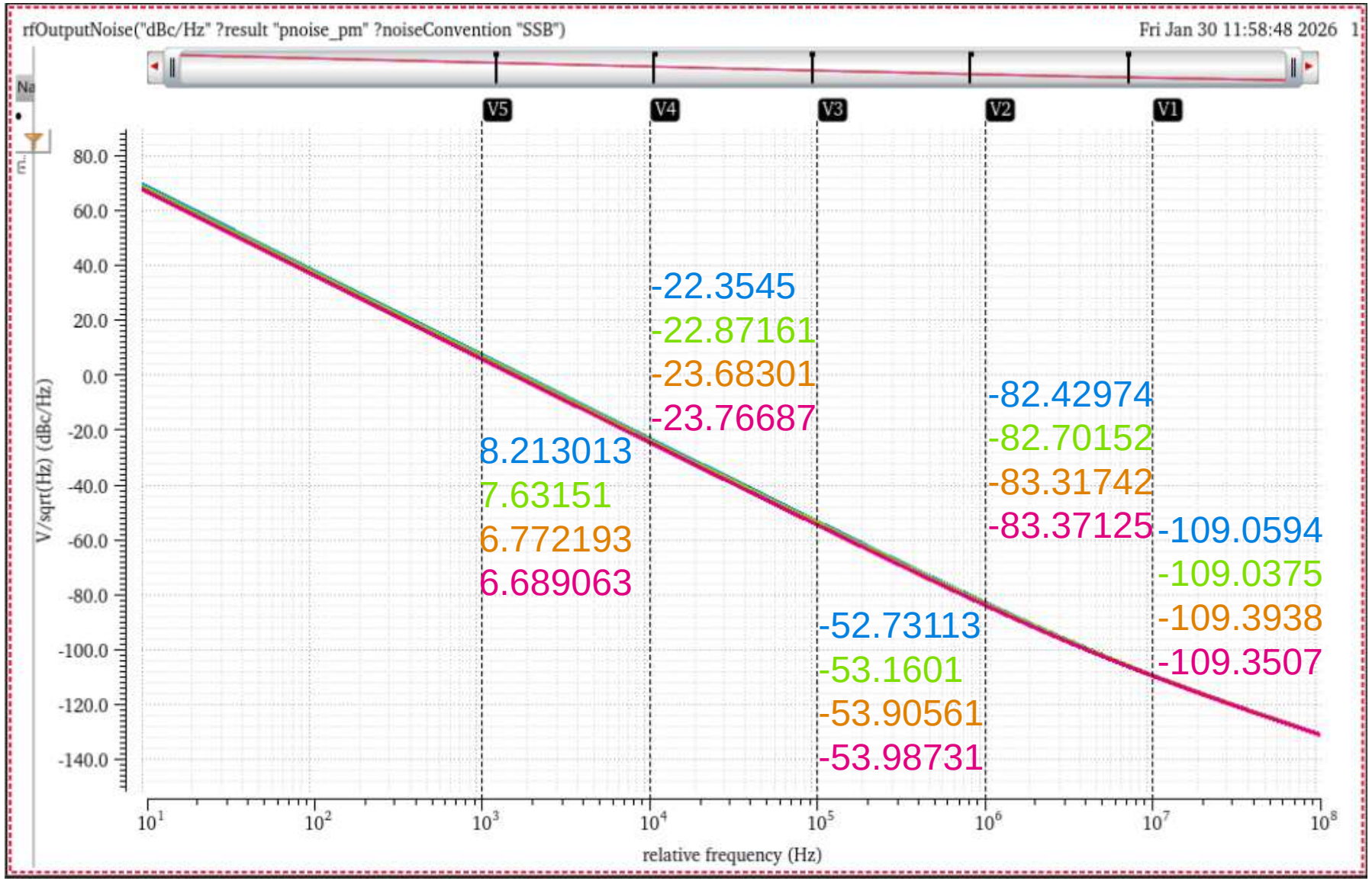
Label needs to write with METAL PIN

PEX



Hz	dB
1k-10k	-30.62 -30.56 -30.50
10k-100k	-30.45 -30.36 -30.28
100k-1M	-30.08 -29.93 -29.78
1M-10M	-28.53 -28.12 -27.64

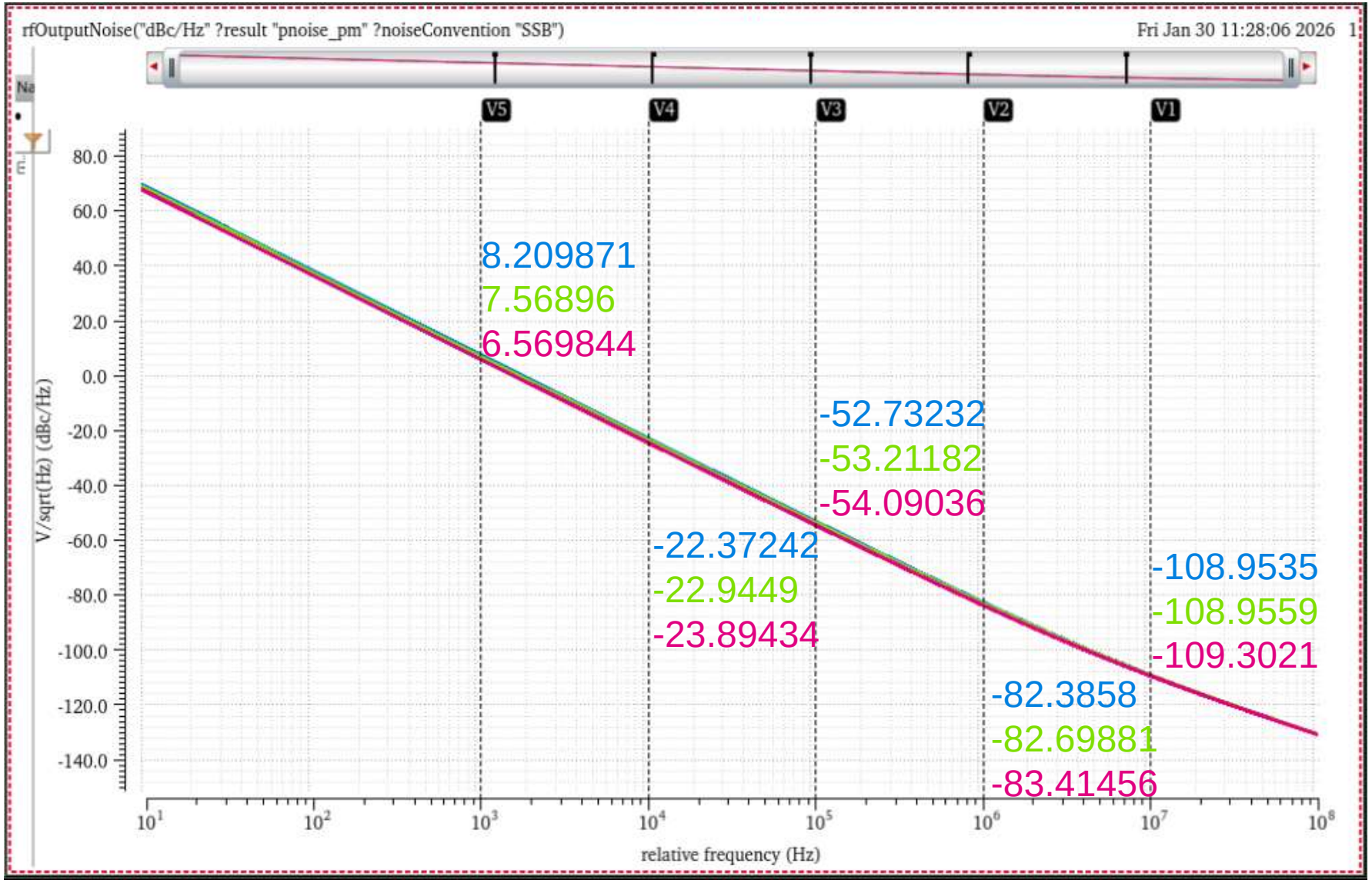
PEX



Hz	dB
1k-10k	-30.57 -30.5 -30.45 -30.45
10k-100k	-30.38 -30.29 -30.23 -30.22
100k-1M	-29.7 -29.54 -29.41 -29.38
1M-10M	-26.63 -26.34 -26.08 -25.98

DSPF

PEX



Hz	dB
1k-10k	-30.62 -30.56 -30.50
10k-100k	-30.45 -30.36 -30.28
100k-1M	-30.08 -29.93 -29.78
1M-10M	-28.53 -28.12 -27.64

Calibreview

PEX

Output	ff	tt	ss
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")			
fosc	26.23G	20.9G	16.31G
idc	5.736m	4.658m	3.719m
pdc/1m	6.883	5.589	4.463
|PN|	74.08	75.32	76.95

ff	tt	ss
ff	tt	ss
-40	40	125

Pre-sim

Output	ff	tt	ss_0	ss_1
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	12.69G	10.73G	9.052G	8.903G
idc	4.089m	3.457m	2.885m	2.879m
pdc/1m	4.907	4.148	3.462	3.455
|PN|	82.43	82.7	83.32	83.37

ff	tt	ss_0	ss_1
ff	tt	ss	ss
-40	40	100	125

DSPF

Output	ff	tt	ss
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")			
fosc	12.72G	10.8G	8.919G
idc	4.006m	3.426m	2.861m
pdc/1m	4.808	4.111	3.433
|PN|	82.39	82.7	83.41

ff	tt	ss
ff	tt	ss
-40	40	125

Calibreview

RCC

PEX

Output	ff	tt	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	18.11G	14.49G	11.64G	11.37G
idc	5.652m	4.564m	3.669m	3.621m
pdc/1m	6.782	5.477	4.403	4.346
|PN|	77.06	78.36	79.82	79.95

C+CC

Output	ff	tt	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	12.69G	10.73G	9.052G	8.903G
idc	4.089m	3.457m	2.885m	2.879m
pdc/1m	4.907	4.148	3.462	3.455
|PN|	82.43	82.7	83.32	83.37

RC
C

Output	ff	tt	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	26.29G	20.97G	16.82G	16.38G
idc	5.667m	4.599m	3.725m	3.669m
pdc/1m	6.8	5.519	4.47	4.402
|PN|	73.82	75.15	76.6	76.77

NoR
C

DSPF

PEX

Output	ff	tt	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	18.21G	15.4G	13G	12.75G
idc	4.076m	3.447m	2.881m	2.883m
pdc/1m	4.891	4.137	3.457	3.46
|PN|	79.35	79.58	80.17	80.24

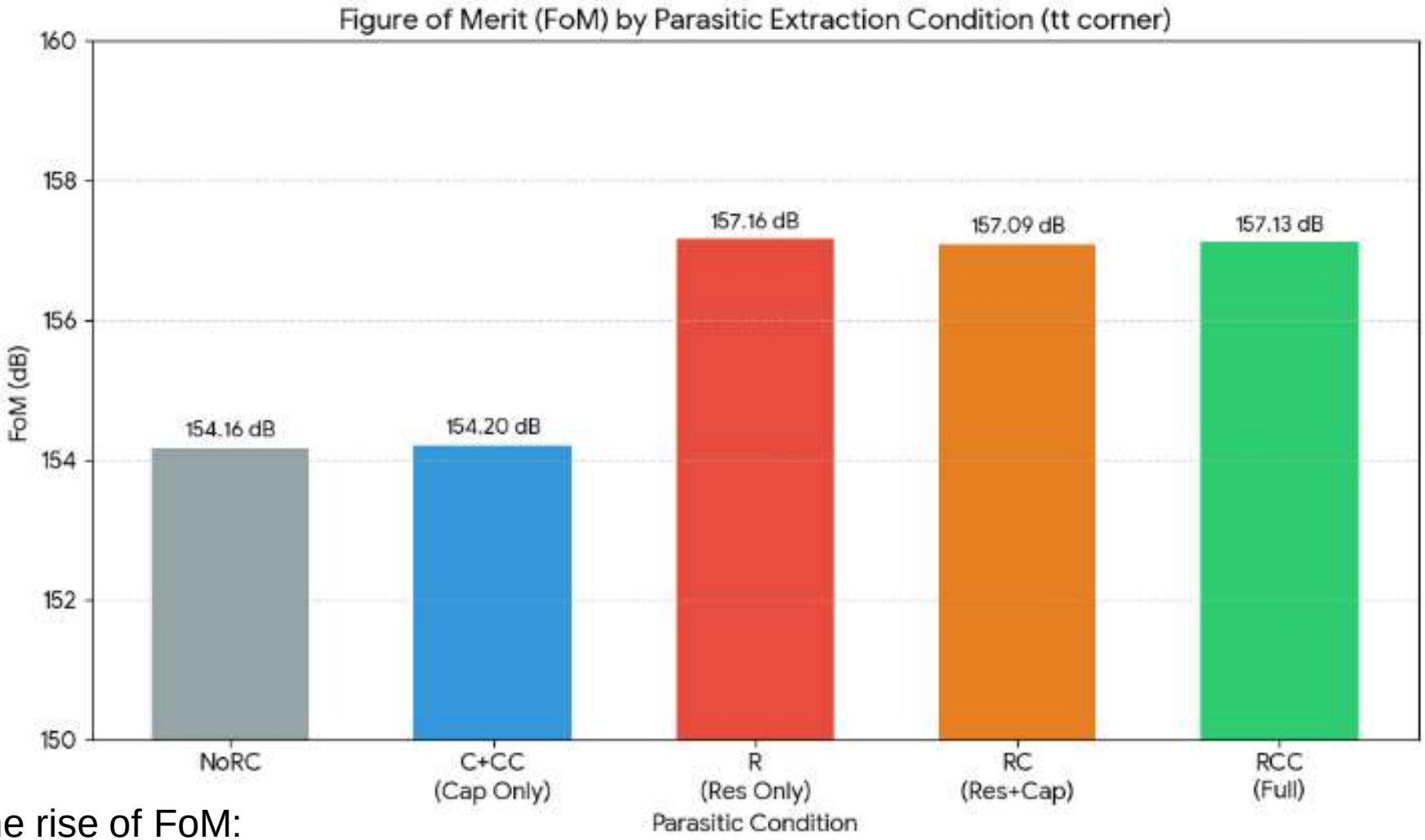
R

Output	ff	tt	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter
rfOutputNoise("dBc/Hz" ?result "pnoise_pm" ?noiseConvention "SSB")				
fosc	13.53G	11.45G	9.647G	9.484G
idc	4.085m	3.453m	2.883m	2.876m
pdc/1m	4.902	4.143	3.46	3.451
|PN|	81.78	82.09	82.73	82.79

RC

DSPF

Comparison

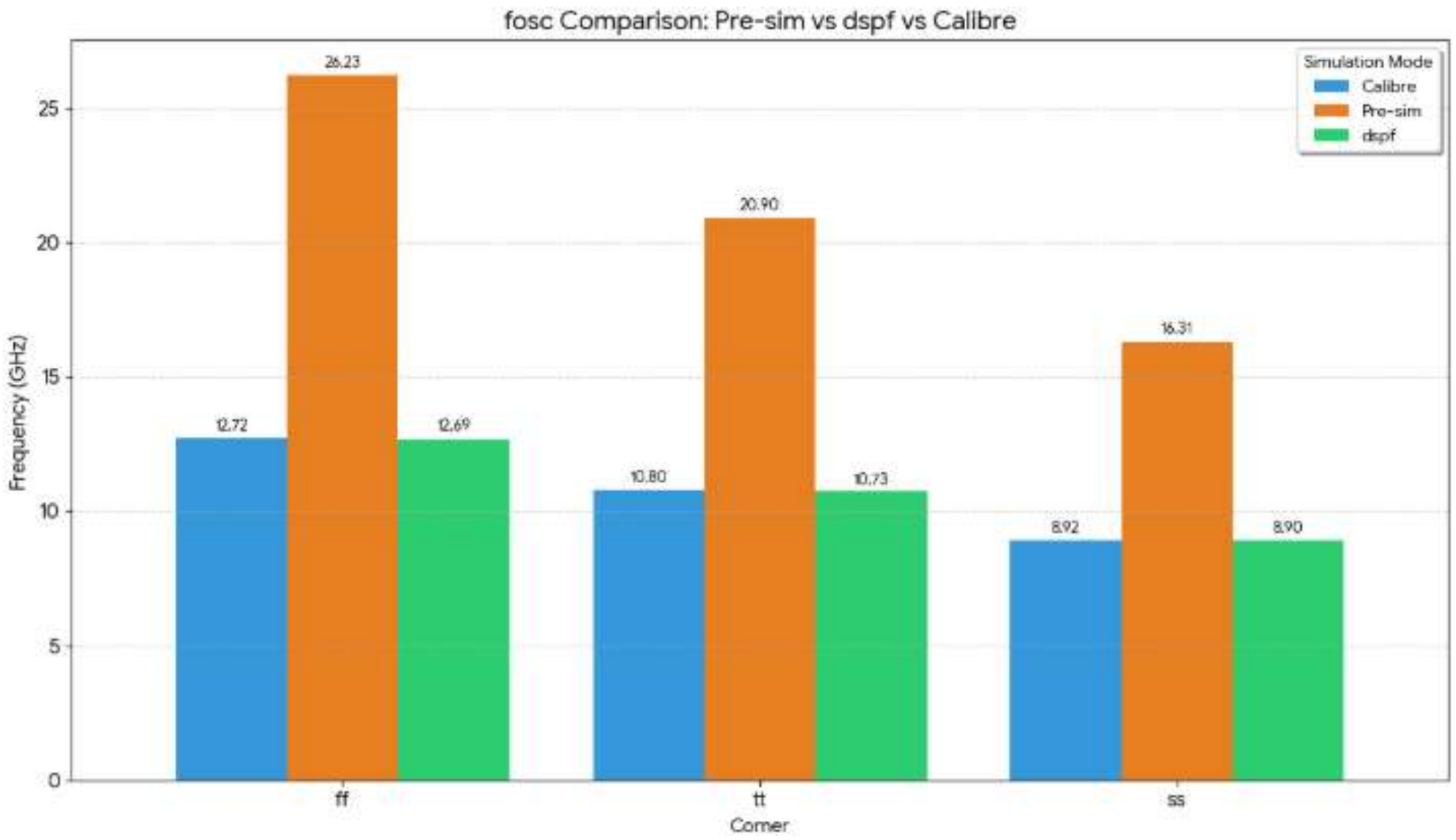


Reason of the rise of FoM:

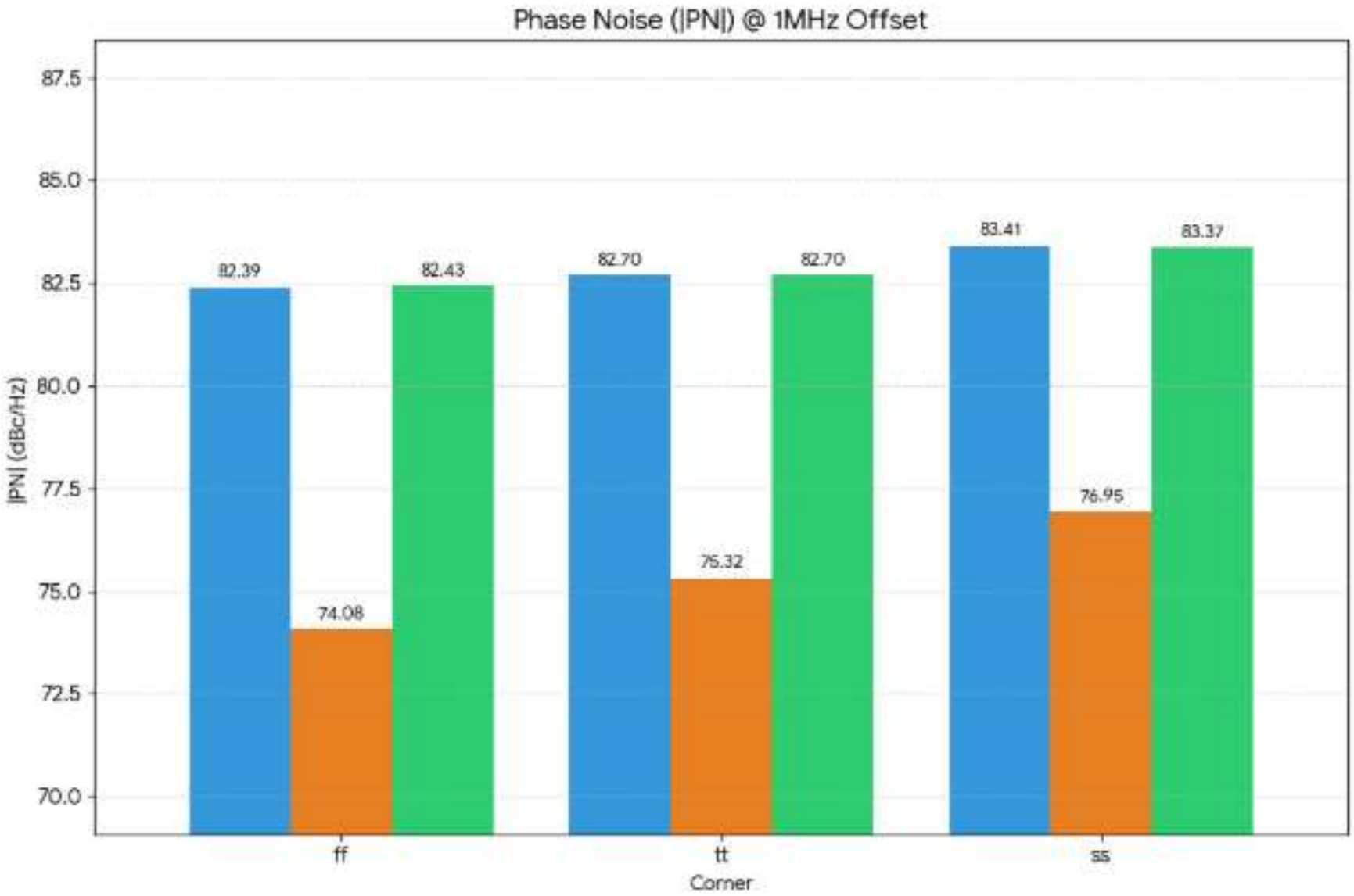
Not because of the performance improvement,

=> the voltage drop caused by the parasitic resistor has abnormally reduced the power consumption.

Comparison


fosc Comparison: Pre-sim vs dspf vs Calibre
Simulation Mode
Calibre
Pre-sim
dspf
Frequency (GHz)
0
5
10
15
20
25
12.72
26.23
12.69
10.80
20.90
10.73
8.92
16.31
8.90
ff
tt
ss
Corner

Comparison


Phase Noise (|PN|) @ 1MHz Offset
87.5
85.0
82.5
80.0
77.5
75.0
72.5
70.0
|PN| (dBc/Hz)
82.39
74.08
82.43
82.70
75.32
82.70
83.41
76.95
83.37
ff
tt
ss
Corner

Comparison

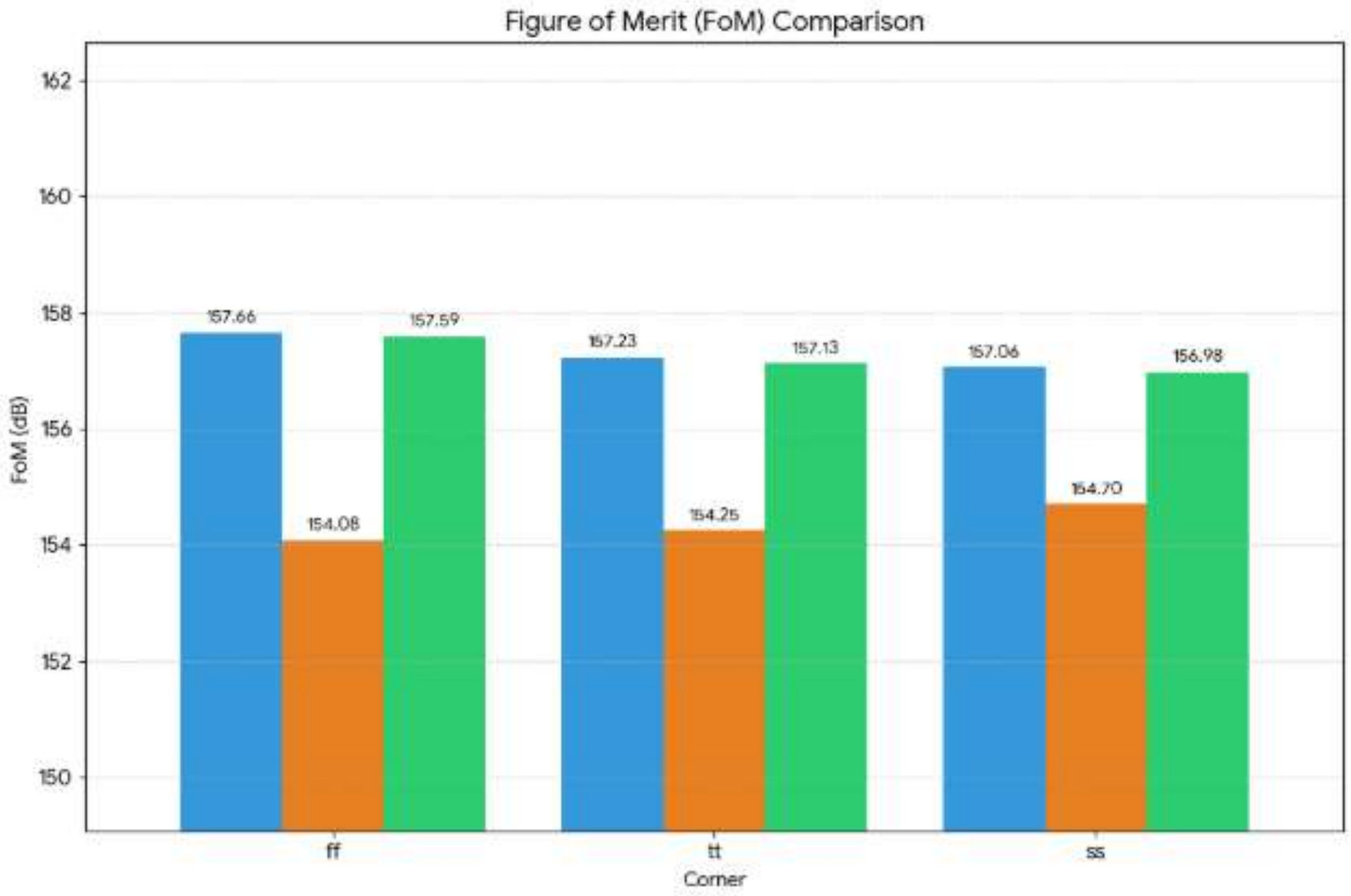


Figure of Merit (FoM) Comparison
FoM (dB)
162
160
158
156
154
152
150
157.66
154.08
157.59
157.23
154.25
157.13
157.06
154.70
156.98
ff
tt
ss
Corner

Summary

◆ The frequency drop was caused by parasitic C and R.

20.9G -> 14.5G by C

- layout wiring, metal fill, parasitic caps around inductors, etc.

20.9G -> 15.4G by R

- High resistance of Power/Ground wiring.

◆ IR Drop by parasitic resistance

- When extracting only parasitic R, the current I_{dc} plummeted by about 26% to 5.6mA -> 4.1mA.
- The wiring from the VDD to the transistor was too thin or long, causing severe voltage drop.

  -> This causes the transistor's bias point to be distorted,

  disrupting normal oscillation and eating away at the frequency.

◆ Calibre view VS DSPF

there is a slight difference between the two extraction methods

- f_{osc}: The Calibre results were 0.2~0.6% higher than the DSPF.
- Phase Noise: almost identical, less than 0.05%.

DSPF is preferred by industry because of portability and simulation tool independence.